PROJECT PORTFOLIO

SMARTSOLVE AI

AI-Powered Complaint Management System

A student project demonstrating how machine learning can help classify civic and service complaints, recommend a department, and support complaint tracking.

DEVELOPED BY
JAYMEET PATEL

Academic Project Portfolio • 2026

School / Institution: _____

Class / Course: _____ **Roll No.:** _____

Submitted to: _____

1. Project Overview

SmartSolve AI is a web-based complaint management prototype. It is designed to let a user describe a civic or service-related issue, receive an automatically predicted complaint category and priority, view a suggested department, and track the complaint using a unique tracking ID.

The application demonstrates a possible way to organize incoming complaints and make them easier to review. It is a college demonstration prototype and is not an official government portal.

2. Problem Statement

People may find it difficult to identify the correct department for an everyday civic or service issue. Complaint information can also be difficult to organize and follow up on when there is no clear tracking reference. SmartSolve AI addresses this project problem by combining a guided complaint form, machine-learning-based category prediction, priority indication, department suggestions, and status tracking.

3. Project Objectives

- Provide a simple online form for submitting complaints.
- Use a machine-learning model to predict a complaint category from its description.
- Indicate a priority level to help distinguish more urgent submissions.
- Suggest a relevant department or helpdesk.
- Generate a tracking ID so users can check complaint status.
- Provide an administrative dashboard to review and manage complaint records.

4. Technologies and Tools

Technology	Role in the project
Python / Flask	Backend application and page routing.
HTML	Structure of the web pages and forms.
CSS	Layout, colors, spacing, and responsive presentation.
JavaScript	Browser-side interactions where required.
Machine Learning	Predicts a complaint category from submitted text.
SQLite	Stores complaint and application records.

5. How the System Works

Step 1 — Describe: The user enters their name, email, location, and a description of the issue.
Step 2 — Analyze: The machine-learning component analyzes the complaint text and predicts a category. The application also assigns a priority indication.
Step 3 — Recommend: The result page displays the predicted category, suggested department, confidence score, and a suggested next step.
Step 4 — Track: The application provides a unique tracking ID that can be entered on the tracking page to view the complaint record and its current status.
Step 5 — Admin review: An administrator can view complaint statistics and complaint records through the dashboard.

6. Main Features

Complaint submission: A guided form collects the details needed to register an issue.
AI category prediction: A text-based model predicts the likely complaint category.
Priority indication: The result page shows a priority label such as Urgent.
Department recommendation: A suggested department is shown to guide the next step.
Complaint tracking: A tracking ID lets a user look up the submitted complaint.
Admin dashboard: Summary cards, charts, filters, and a complaint list support administrative review.

7. Machine Learning Component

The application uses a machine-learning approach to interpret the text entered by the user and predict a complaint category. The model's prediction is displayed with a confidence score. In the sample result captured for this portfolio, the predicted category is **Other** and the displayed confidence score is **27.1%**. This is one example output, not a claim of overall model accuracy.

A confidence score should be interpreted as the model's confidence for that particular prediction; it is not the same as a verified accuracy percentage across a test dataset.

8. Application Screenshots

The following screenshots are from the working SmartSolve AI application supplied for this portfolio.

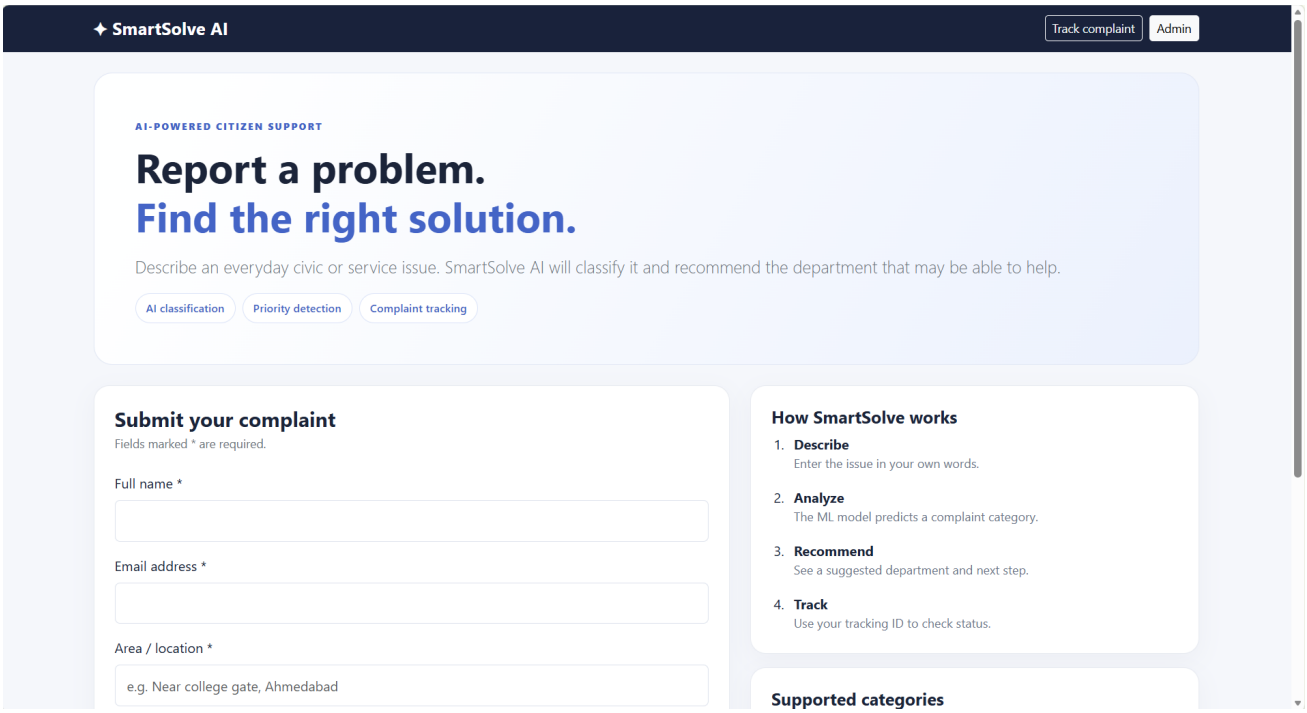


Figure 1. Home page and complaint submission form.

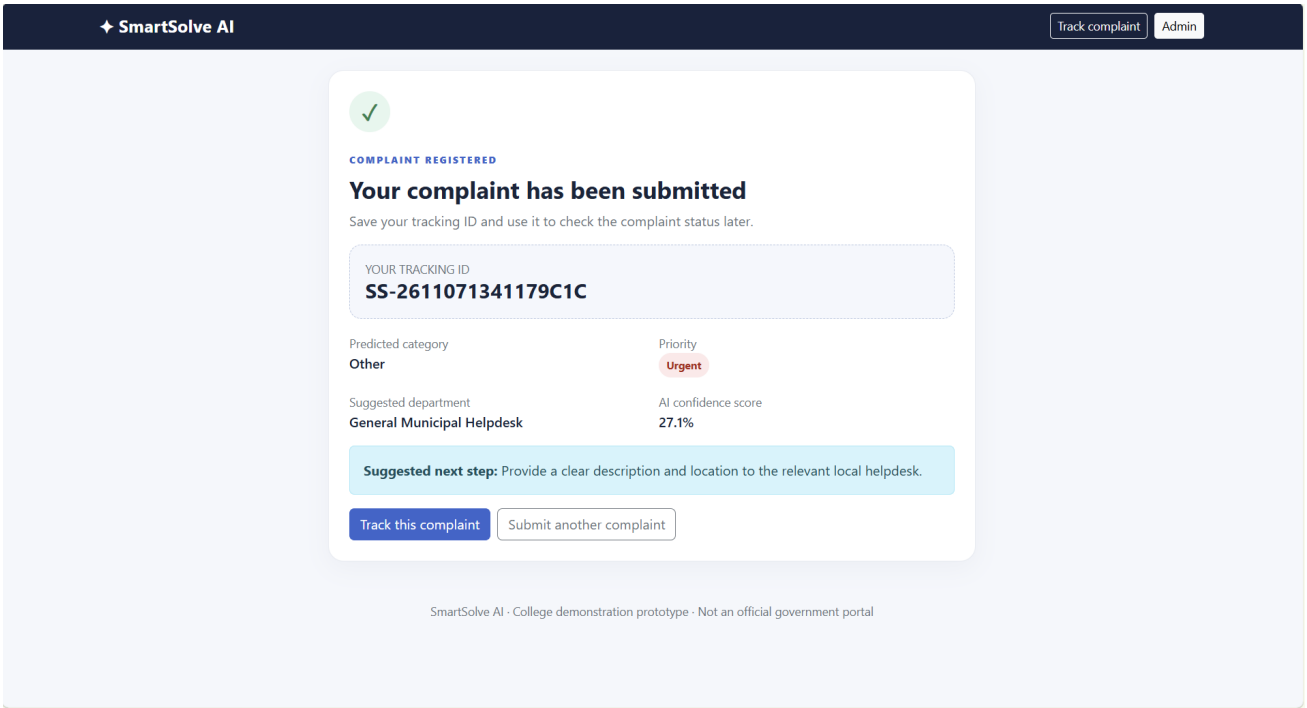


Figure 2. Complaint registration confirmation, tracking ID, predicted category, priority, and suggested department.

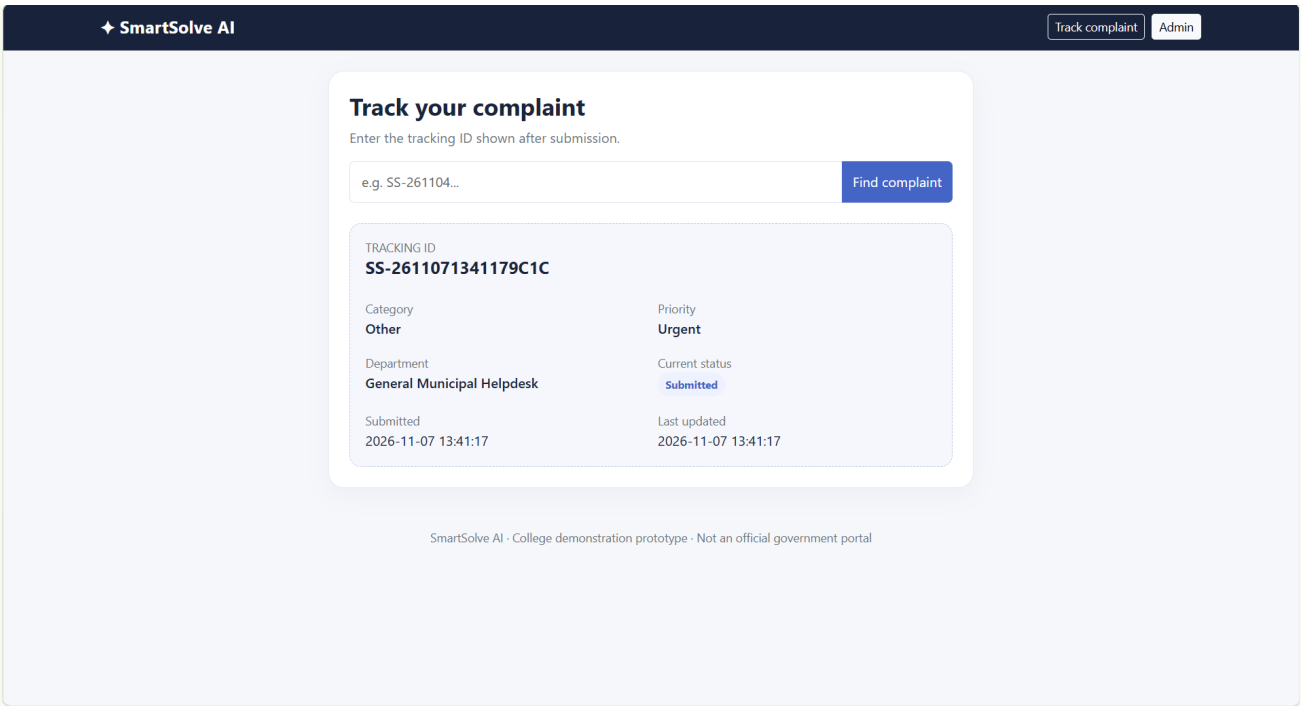


Figure 3. Complaint tracking page showing the record and current status.

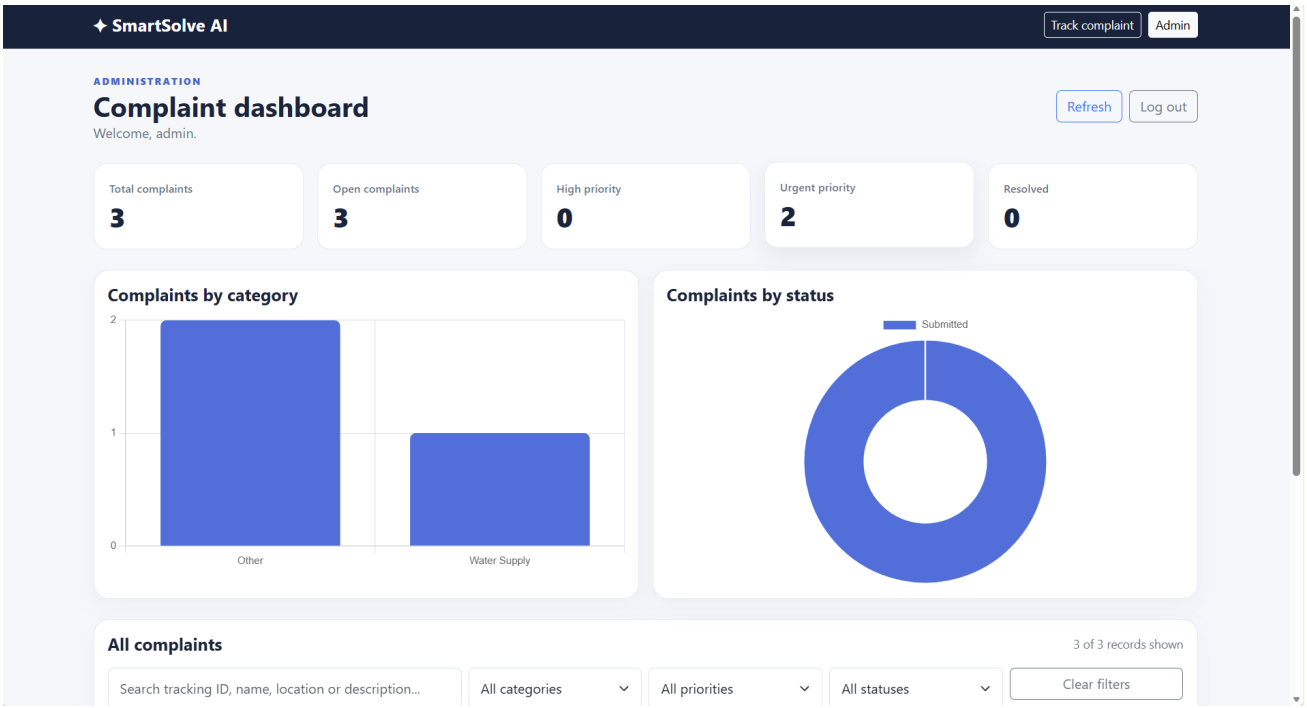


Figure 4. Administrative dashboard with summary statistics, category chart, status chart, and complaint records.

9. Sample Testing and Observed Output

The supplied screenshots show a successful complaint registration flow and a subsequent tracking lookup. The sample record displays a generated tracking ID, predicted category, priority, suggested department, confidence score, and submitted status.

Test area	Observed evidence in screenshots
Complaint submission	A complaint confirmation page is displayed after submission.
Prediction output	Category: Other; priority: Urgent; confidence: 27.1% in the captured example.
Department suggestion	General Municipal Helpdesk is displayed for the sample record.
Tracking	The tracking page retrieves the sample record and displays status: Submitted.
Admin dashboard	The dashboard displays 3 total complaints, 3 open complaints, 2 urgent-priority complaints, and 0 resolved in the captured view.

10. Advantages

• Makes complaint submission straightforward.
• Provides a consistent tracking reference for follow-up.
• Uses automated text classification to assist complaint organization.
• Presents complaint summaries visually for administrative review.
• Demonstrates how web development, databases, and machine learning can be combined in one application.

11. Limitations

• Predictions depend on the quality and coverage of the training data.
• A low-confidence or incorrect category may require human review.
• Department suggestions are guidance within this prototype and do not guarantee official action.
• The captured dashboard represents sample project data, not real public-service statistics.
• The application is a college demonstration prototype, not an official government service.

12. Future Scope

Possible improvements include expanding and balancing the training dataset, evaluating the model on a separate test set, adding multilingual complaint support, enabling notifications for status changes, improving accessibility, and integrating with verified department workflows if appropriate permissions and partnerships are available.

13. Conclusion and Learning Outcomes

SmartSolve AI demonstrates a complete complaint-management workflow, from submitting an issue to receiving a prediction and tracking its status. Developing the project provides practical experience in designing web pages, handling form data, connecting a backend to a database, applying machine learning to text, and presenting information through an administrative dashboard.

The project also highlights the importance of testing AI predictions and keeping a human review process for real-world decisions.

— END OF PROJECT PORTFOLIO —

Prepared by: Jaymeet Patel
Project: SmartSolve AI
Academic year: 2026